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REMANENT MAGNETIZATION AND SUSCEPTIBILITY OF LATE CENOZOIC ROCKS FROM NEW ZEALAND*

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ABSTRACT

Measurements of magnetic remanence and susceptibility were made on 176 samples of volcanic rocks from 22 late Cenozoic formations from the North Island. Intensities of remanence, susceptibilities, and Q ratios are highly variable in both the ignimbrites and andesites and their ranges overlap, indicating that the magnetic anomalies produced by ignimbrite bodies would be difficult to distinguish from those produced by andesite flows. An analysis of the directions of magnetization of the ignimbrites indicates that groups of petrologically distinct ignimbrites probably formed during the same volcanic episode.

INTRODUCTION

Late Cenozoic volcanic formations on North Island assigned to the Hawera Series are normally magnetized and therefore probably were formed during the Brunhes normal polarity epoch (Cox, 1969a). This epoch, during which the earth's field was directed toward the north, began 0·69 m.y. ago and continues in the present (Cox, 1969b). Almost all of the ignimbrite deposits which fill the Taupo graben are normally magnetized and undoubtedly were erupted during this short interval of time. These include all the ignimbrites assigned to the Hawera Series in the Taupo region (Grindley, 1960) and Rotorua region (Healy *et al.*, 1964) and all but two of the ignimbrites assigned to the Wanganui Series. Sample sites are shown in Fig. 1 and the stratigraphic sequence of the ignimbrites in Table 1. Grid references are given in the Appendix. The two reversely magnetized ignimbrites on the west side of the Taupo graben are the Waitetariki Ignimbrite (Site NZ8), which Healy *et al.* (1964) assign to the Wanganui Series, and an ignimbrite of the Whitianga Group (Site NZ5) which Schofield (1967) assigns to the Wanganui Series or the older Taranaki Series. It should be emphasized that the terms Hawera Series and Wanganui Series as used in the present discussion and in the cited references are time stratigraphic units based on fossil control and not on isotopic dating. Moreover, fossiliferous strata are rare in this volcanic sequence. One of the contributions of the paleomagnetic work to the study of the volcanic stratigraphy of North Island has been to provide an independent time horizon. The change from reversed to normal polarity that occurred 0·69 m.y. ago appears to

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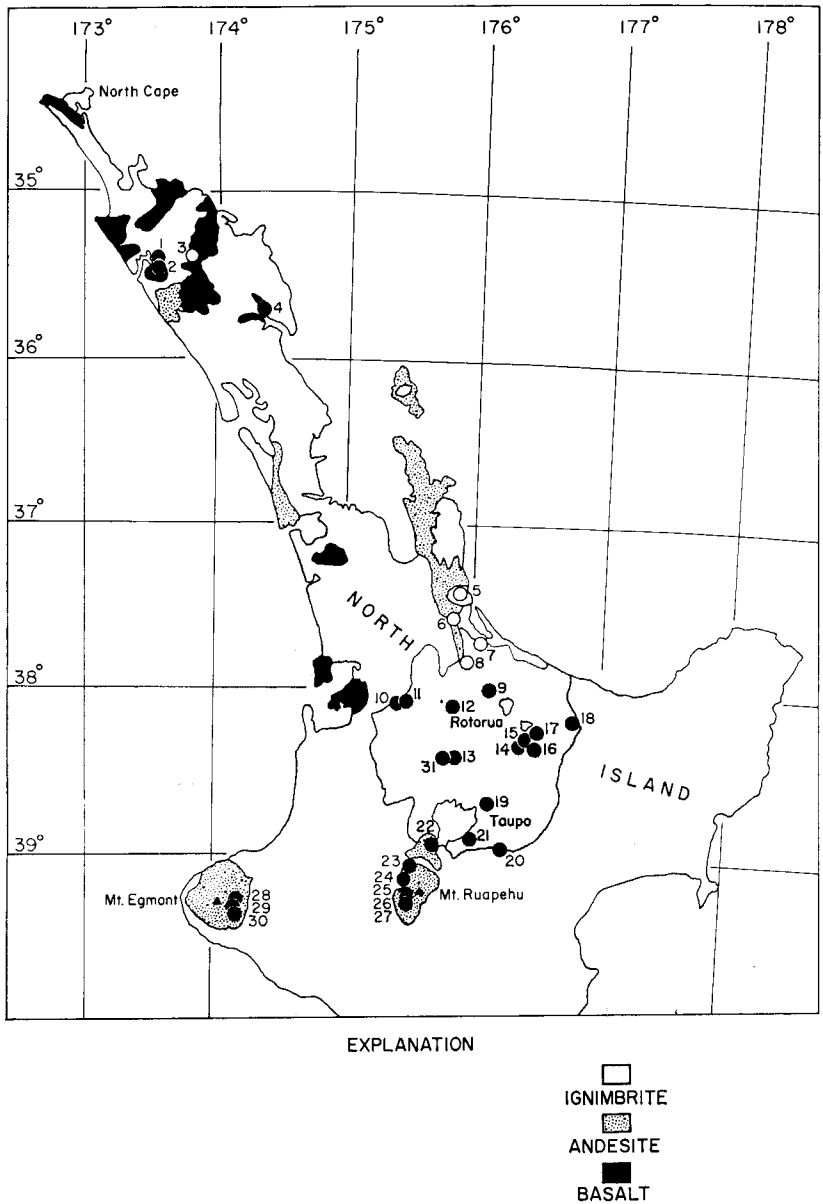


FIG. 1—Paleomagnetic sampling sites on North Island. Site 31 is the Whakamaru Ignimbrite previously reported by Hatherton (1954). Open circles—reversely magnetized sites: black circles—normally magnetized sites.

TABLE 1—Stratigraphy of the Ignimbrites in the Regions of Taupo (Grindley, 1960), Rotorua (Healy *et al.*, 1964), and Auckland (Schofield, 1967)

		Rotorua and Auckland			Taupo		
	Series	Name of ignimbrite formation	N or R	Site	Name of ignimbrite formation	N or R	Site
	Hawera	Kaingaroa	N	NZ16	Kaingaroa		
		Mamaku	N	NZ9			
		Matahina	N	NZ18			
		Mokai					
Brunhes normal polarity epoch	Wanganui	Manunui	N	NZ11			
		Whakamaru	N	NZ31	Rangitaiki	N	NZ21
		Rangitaiki			Whakamaru	N	NZ31
		Waiotapu	N	NZ15	Te Whaiti	N	NZ20
		Marshall	N	NZ12	Wairakei		
		Paeroa					
		Pakaumanu	N	NZ10	Pakaumanu		
0·69 m.y.		Te Kopia	N	NZ14			
Matuyama reversed polarity epoch		Waitearíki	R	NZ8			
		Papamoa					
		Whitianga	R	NZ5			

fall within the interval between the eruptions of the Waitetariki and Te Kopia Ignimbrites (Table 1). Apparently ignimbrite volcanism began on North Island shortly before 0·69 m.y. ago, the beginning of the Brunhes normal polarity epoch.

In an analysis of the ash content and paleomagnetism of deep-sea sediment cores collected east of New Zealand, Ninkovich (1968) distinguished five rhyolitic ash layers which he correlated with major ignimbritic eruptions on New Zealand. The age of the oldest ash bed based on the paleomagnetic stratigraphy of the sediments was 0·86 m.y. Both studies thus indicate that the ignimbrite eruptions began shortly before the last change to normal polarity.

The present report is an extension of the earlier work with two objectives in view. The first is to determine whether the magnetic remanence and susceptibility of different rock types are sufficiently distinctive to provide a basis for interpreting magnetic anomaly maps. The second is to subdivide the ignimbrites that formed during the Brunhes normal polarity epoch into several stratigraphic groups on the basis of small but significant variations in their directions of magnetization.

Intensity of Remanence and Susceptibility

At 22 sites, 8 unweathered oriented samples were cored from the outcrops using a portable, water-cooled diamond drill. The remanence was measured using a spinner magnetometer with an accuracy of 5% (Doell and Cox, 1967). The mean intensity of the original natural remanence of the eight samples from each site is given in Table 2. At eight other sites shown in Fig. 1, measurements were made of the magnetic polarity of four or more samples at the outcrop using a portable magnetometer and one sample was retained for a check of the polarity in the laboratory. However, because of the large intrasite variations which characterize volcanic formations (Hatherton, 1954; Hoare, 1967), the data from these eight sites were not used in the statistical analysis. The means were calculated in two ways, as simple averages and as logarithmic averages. Irving *et al.* (1966) have shown that the intensities of remanence and the susceptibilities of many groups of rocks do not have normal distributions but may be described with a better fit by a log-normal distribution. The best estimate of the mean is therefore given by the antilogarithm of the mean of the logs of the intensities or susceptibilities. As expected, it was found that where the standard deviations are small, as at Site NZ9, the difference between the mean and the logarithmic mean is small. At sites with large variances such as NZ27, however, the log mean is significantly smaller than the mean and provides a better measure of central tendency.

Extreme variations in the intensity of remanence characterize these rocks, on both the within-site and between-site levels. For example, the andesite samples from Site NZ29 have intensities varying by a factor of 5·7. The samples from this site with larger values of remanence have smaller values of susceptibility (Fig. 2), suggesting variations in the type of ferromagnetic mineral present and not simply variations in the amount. The most likely

TABLE 2—Values of Original Remanence, Susceptibility, and Q Ratio. Each value is based on measurements of eight samples from one site.

Site no.	Remanence (× 1000 emu/cc)			Susceptibility (× 1000 emu/cc)			Q (F = 0·57 oe.)		
	Mean	S.D.	Log. mean	Mean	S.D.	Log. mean	Mean	S.D.	Log. mean
				Ignimbrites					
NZ9	0·265	0·073	0·257	0·739	0·057	0·737	0·636	0·190	0·611
NZ11	0·161	0·027	0·150	0·244	0·027	0·242	1·15	0·46	1·08
NZ12	1·41	0·25	1·39	0·635	0·032	0·634	3·90	0·66	3·85
NZ15	1·26	0·34	1·21	0·865	0·156	0·850	2·53	0·38	2·50
NZ16	4·17	1·19	4·06	0·648	0·040	0·646	11·3	2·99	11·0
NZ18	1·39	0·27	1·37	0·618	0·045	0·616	3·96	0·73	3·89
NZ20	1·30	0·13	1·30	0·959	0·045	0·958	2·40	0·33	2·38
NZ21	9·25	1·33	9·18	0·784	0·009	0·784	20·7	2·88	20·5
				Andesites					
NZ22	0·744	0·313	0·692	1·55	0·26	1·53	0·895	0·496	0·794
NZ23	0·704	0·154	0·690	0·365	0·085	0·356	3·53	1·04	3·41
NZ24	4·78	0·79	4·72	0·441	0·086	0·432	19·2	1·66	19·1
NZ25	1·19	0·31	1·15	4·24	0·30	4·23	0·493	0·132	0·476
NZ26	1·16	0·34	1·11	3·03	0·27	3·02	0·678	0·225	0·643
NZ27	4·39	2·83	3·71	1·85	1·55	1·53	6·61	6·68	4·24
NZ28	11·6	0·6	11·6	9·25	0·56	9·24	2·21	0·16	2·21
NZ29	2·68	1·86	2·23	4·45	1·26	4·22	1·53	2·00	0·93
NZ30	4·59	1·88	4·25	6·54	1·38	6·41	1·38	0·80	1·16
				Basalts					
NZ2	4·32	1·80	3·88	1·03	0·11	1·03	7·48	3·40	6·61
NZ4	1·72	3·06	0·65	0·152	0·093	0·129	29·7	65·2	8·8
				Rhyolites					
NZ13	0·195	0·035	0·192	0·294	0·113	0·273	1·28	0·36	1·23
NZ17	0·153	0·028	0·150	0·251	0·054	0·246	1·10	0·23	1·08
				Dacite					
NZ19	0·611	0·059	0·608	1·04	0·11	1·03	1·04	0·11	1·04

Site numbers explained in Appendix

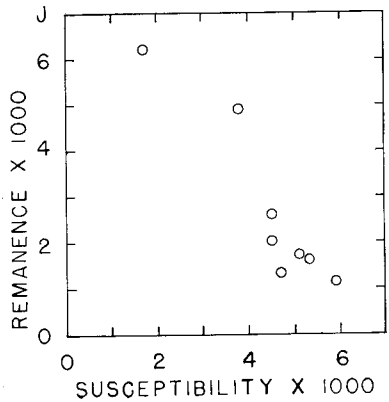


FIG. 2—Intensity of original remanence and susceptibility of eight samples from Site NZ29, an andesite flow of Mount Egmont.

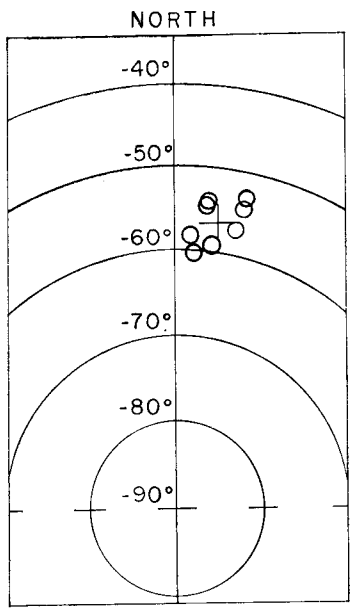


FIG. 3—Directions of magnetization of the eight samples from Site NZ29 after partial demagnetization in an alternating field of 100 oersteds. All points are on the upper hemisphere of an equal area projection. The mean is shown by a cross.

explanation is that the samples with low susceptibility and large remanence underwent more oxidation during the late stages of the cooling of the andesite flow. Another possibility is that varying amounts of oxidation occurred during weathering. However, this is unlikely because the directions of magnetization of these samples are tightly grouped after magnetic cleaning in a moderate alternating field of 100 oersteds to remove small amounts of unstable magnetization due to lightning and viscous effects (Fig. 3). If the ferromagnetic minerals had undergone oxidation through weathering over a long period of time, part of the stable remanence of the samples would be chemical remanence and the directions of this remanence would be dispersed, reflecting changes in the direction of the earth's field that occurred after the flow cooled.

On the between-site level, variations in remanence are even greater. The Rangitaiki Ignimbrite (Site NZ21) has a mean remanence 61 times larger than that of the Manunui Ignimbrite (Site NZ11). The difference is probably not due to variations in the intensity of the ancient field because, for reasons discussed below, the two ignimbrites appear to be of about the same age. Part of the difference is probably due to differences in magnetite content, as suggested by the four-fold difference in susceptibility. The rest of the difference is probably due to variations in the capability of the magnetite to acquire thermoremanent magnetization, a property very sensitive to the grain size, the dislocation density, and the titanium content of the ferromagnetic minerals present. Hatherton (1954) and Hoare (1967) showed that even within one ignimbrite body, the intensity of remanence may vary by a factor of 10 or more.

The site mean intensities of the andesites and ignimbrites have ranges that are broad (Fig. 4) and that overlap. The overall averages for the two groups (Table 3) are not significantly different at the 95% confidence level.

The susceptibility of the eight samples collected from each site was measured in an oscillating magnetic field with a peak amplitude of 1 oersted and a frequency of 10^3 HZ. The accuracy of the measurements was about 10%. The mean, the standard deviation, and the log mean for each site are listed in Table 2. The within-site standard deviations of susceptibility are much smaller than those of remanence, both for the andesites and the ignimbrites. This reflects the relative insensitivity of susceptibility to the structure-sensitive properties that control remanence On the between-site level, the ignimbrites, with exception of Site NZ11, have remarkably similar susceptibilities. This suggests that their content of titanomagnetite is similar. In contrast, susceptibilities are highly variable from site to site in the andesites, suggesting that the content of titanomagnetite is also variable.

The Koenigsberger or Q ratio is defined by

$$Q = J/F\chi \qquad (1)$$

where J is the remanence, F is the intensity of the field at the sampling site, and χ is the apparent volume susceptibility. The Q ratio is particularly important in interpreting magnetic anomalies. The total magnetization of the rocks producing the anomalies consists of two vector components, an induced component with intensity $F\chi$ parallel to the earth's field, and a remanent component with intensity J which is probably parallel to the field

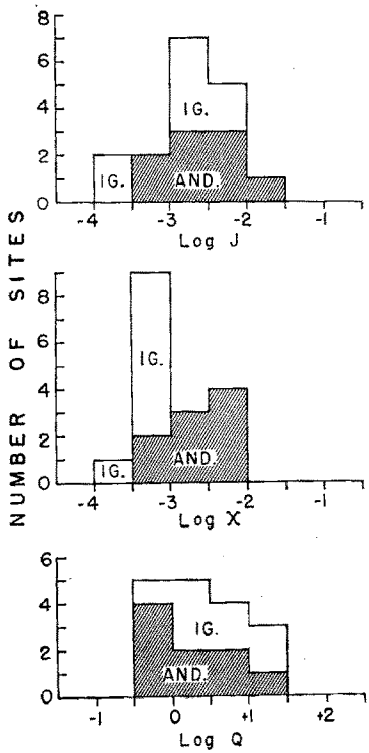


FIG. 4—Histogram of the remanence J, susceptibility χ, and Koenigsberger ratio Q of the ignimbrites (IG.) and andesites (AND.). Each entry is the logarithmic mean of measurements on eight samples from one site. $Q = J/0{\cdot}57\,\chi$.

that existed when the rock formed but which is not necessarily parallel to the present field. At the latitude of New Zealand, rock bodies with small values of Q produce anomalies that predominantly add to the regional field above the body and thus, by definition, are positive. In contrast, rocks with large values of Q may produce either negative or positive anomalies, depending on whether the remanence is normal or reversed. As most of the ignimbrites have values of $Q \geqslant 2$, the anomalies associated with them will reflect the polarity of their remanence. The andesites are about equally divided between values of $Q < 1$ and $Q > 1$. Unfortunately, the Q values of the ignimbrites and andesites overlap, as do the values of J and χ. Thus there is nothing in the present set of results to suggest that bodies of either type would possess a unique "magnetic signature" recognizable on magnetic anomaly maps unless, as is quite likely, the andesite and ignimbrite bodies possess different characteristic shapes.

DIRECTIONS OF MAGNETIZATION OF THE IGNIMBRITES

The possibility that all of the ignimbrites on North Island formed during five distinct pulses was suggested by the discovery of Ninkovich (1968) that deep-sea sediments near New Zealand contain five distinct layers of rhyolite ash. Two of the ash layers were deposited before the Brunhes epoch.

TABLE 3—Overall Means, Standard Deviations, and Log Means of the Remanence and Susceptibility of the Ignimbrites and Andesites. The input data were the log. means from Table 2.

Petrologic type	No. of sites	Remanence (× 1000)			Susceptibility (× 1000)			Q (F = 0·57 oe.)		
		Mean	S.D.	Log. mean	Mean	S.D.	Log. mean	Mean	S.D.	Log. mean
Ignimbrites	8	2·36	3·00	1·20	0·683	0·214	0·642	5·73	6·79	3·28
Andesites	9	3·35	3·47	2·19	3·44	2·95	2·19	3·67	5·96	1·76

Their ages obtained by extrapolation from the known age of the beginning of the Brunhes epoch are 0·86 and 0·73 m.y. Three of the ash layers are within the Brunhes normal polarity epoch and have interpolated ages of 0·67, 0·31, and 0·27 m.y.

How are these five layers related to the numerous ignimbrite formations that have been recognized on land? In the Rotorua area, Healy *et al.* (1964) recognize 14 ignimbrite formations, of which 12 are within the Brunhes epoch (Table 1). Of course it is possible that because of variations in wind direction or in the style of volcanism, some of the ignimbrite eruptions may not have deposited ash in the ocean east of New Zealand. On the other hand, if ash layers from all the ignimbrite eruptions are preserved in the deep-sea sediment cores, then presumably some of the lithologically distinct ignimbrites are correlative and the number of distinct eruptive episodes is smaller than the number of ignimbrite formations. Support for this is provided by the paleomagnetic results (Table 4) from the nine ignimbrite formations with ages in the Brunhes epoch.

The basis for this interpretation is as follows. Because of geomagnetic secular variation, the direction of the earth's field changes over a range of several tens of degrees during any one polarity epoch. The rate varies from several degrees per century to several tens of degrees per century. At the present time, the rate varies widely over the earth's surface, and at any one place, the rate varies as a function of time. Therefore, even if the directions of magnetization of two volcanic units are accurately known, it is not possible to calculate a minimum age difference between them. However, the paleomagnetic data serve to limit the age relations between the two units in the following ways. (1) If the sampling sites have different directions of magnetization, they cannot be from the same volcanic cooling unit. (2) If the sampling sites have the same magnetic direction, they may be but are not necessarily from the same cooling unit.

TABLE 4—Directions of Magnetization of Ignimbrites.

Site	N	R	K	α_{95}	Incl.	Decl.	δ
NZ9	8	7·95071	142	4·66	−84·2	118·0	35·7
NZ11	8	7·97807	319	3·10	−65·2	000·6	7·8
NZ12	8	7·99591	1713	1·34	−63·4	343·1	10·2
NZ15	8	7·98537	478	2·54	−65·9	005·2	8·5
NZ16	8	7·98671	527	2·42	−50·6	008·2	9·3
NZ18	8	7·99310	1014	1·74	−49·8	006·7	8·8
NZ20	8	7·99223	901	1·85	−65·9	357·4	7·8
NZ21	8	7·98362	427	2·68	−64·1	357·7	6·0
NZ31	52				−65·3	351·1	

N, number of samples. R, vector sum of N unit vectors parallel to directions of magnetization. K, precision parameter. α_{95}, radius of circle of 95% confidence. Incl., mean inclination. Decl., mean declination. δ, angle between the mean magnetic direction and the direction of an axial dipole field.

The magnetic directions of the ignimbrites are plotted on Fig 5 in the form of 95% confidence circles about the site means. The three shaded circles are for the three sites from the Hawera Series: the Mamaku Ignimbrite (NZ9), the Kaingaroa Ignimbrite (NZ16), and the Matahina Ignimbrite (NZ18). The magnetic directions of the latter two are not significantly different and therefore they may have formed at the same time. Hoare (1967) has reported additional results from two sites in the Matahina Ignimbrite with confidence limits of 21° and 13°. The mean direction for Site NZ18 falls within both circles.

What is the probability that the agreement between the direction of magnetization of Sites NZ16 and NZ18 is a simple coincidence? A quantitative estimate may be made as follows. We assume that over a long period of time, angular deviations of the earth's field from the field of an axial dipole may be described by the function (Fisher, 1953)

$$P(\delta)dA = \frac{\kappa}{4\pi \sinh \kappa} \exp(\kappa \cos \delta)\, dA \qquad (2)$$

where δ is the departure of the field from the direction of the dipole field, κ is an inverse measure of the angular variance of the field, and $P(\delta)dA$ is the probability that at any particular time in the past, the direction of the field was within an element of solid angle dA. Let δ_9, δ_{16}, and δ_{18} be the angles between the direction of the dipole field and the magnetic directions at Sites NZ9, NZ 16, and NZ18, and let α_9, α_{16}, and α_{18} be the corresponding 95% confidence limits. Then given the direction of a hypothetical first flow, say NZ18, the probability that a later flow, say NZ16,

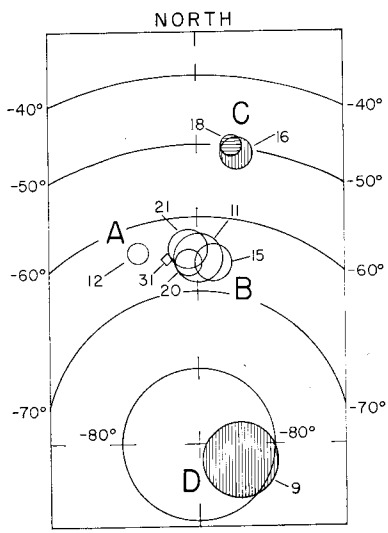


FIG. 5—Directions of magnetization of ignimbrites from the Brunhes normal polarity epoch plotted on the upper hemisphere of an equal area projection. The circles are 95% confidence limits. The three shaded circles are the three sites (NZ9, NZ16, and NZ18) from the Hawera Series. The open circles and open square are six sites from the Wanganui Series and fall into two groups, "A" with one site and "B" with five sites. Confidence limits are not known for Site NZ31 (Hatherton, 1954). The four magnetic groups A, B, C, and D define four successive directions of the geomagnetic field in New Zealand.

will form at a time when the field happens to have the same direction within the limits established by the two 95% confidence circles α_{16} and α_{18} is approximately

$$P_{16,18} = \frac{\kappa}{2\pi}\left(1 - \cos(\alpha_{16} + \alpha_{18})\right)\exp\kappa(\cos\delta_{18} - 1) \qquad (3)$$

On North Island, the value of the parameter κ that describes the average angular dispersion of the earth's field about the axial dipole field is $\kappa = 38$. Using the other data from Table 4, the value of the desired probability is $P_{16,18} = 0{\cdot}01$.

There was no *a priori* reason for assuming that Site NZ16 would have the same magnetic direction as NZ18, and that site NZ9 would not. Therefore it is also necessary to calculate the probability that a site with the confidence limit of NZ9 would randomly have the same direction as Site NZ18. The data in Table 4 give a value of $P_{9,18} = 0{\cdot}024$. Given the direction of Site NZ18, the probability $P_{16,18} + P_{9,18}$ that either Site NZ16 or Site NZ9 or both would have the same direction as NZ18, if their ages were all different, is only 0·034.

The paleomagnetic directions thus indicate that the Matahina Ignimbrite (NZ18) and the Kaingaroa Ignimbrite (NZ16) may have formed at the same time or in rapid succession and that the Mamaku Ignimbrite (NZ9) formed at a different time. The available geologic evidence is as follows (Healy *et al.*, 1964, and Healy, pers. comm.). The two ignimbrites that are petrologically the most similar are the Matahina and the Mamaku, which differ mainly in the relative amounts of hypersthene and hornblende they contain. Both units do not outcrop at any one locality, but they appear to occupy nearly the same stratigraphic position. The Kaingaroa Ignimbrite clearly overlies the Matahina Ignimbrite and is younger, although how much younger is uncertain. These two observations suggest that the Matahina and the Mamaku are about the same age, whereas the Kaingaroa is younger. In disagreement with this conclusion is the observation that the Kaingaroa Ignimbrite has undergone extensive erosion, whereas the Mamaku Ignimbrite is moderately well preserved, suggesting that the Kaingaroa and the Matahina are the two older units and that the Mamaku is considerably younger. Thus the available geologic evidence is split. The interpretation based on relative degrees of erosion is consistent with the paleomagnetic results; the interpretation based on petrologic correlation is not.

Turning to the ignimbrites of the Wanganui Series, four of the sites with known confidence circles have directions of magnetization that are not significantly different from each other. These are designated group B in Fig. 5. Assuming that the Te Whaiti Ignimbrite (Site NZ20) is the oldest, the probability that three of the remaining four ignimbrites would randomly have the same direction as Site NZ20 is given by

$$P = P_{12,20}\,P_{15,20}\,P_{21,20} + P_{11,20}\,P_{15,20}\,P_{21,20} + P_{11,20}\,P_{12,20}\,P_{15,20}$$
$$= 6 \times 10^{-6}.$$

The directions of magnetization of the ignimbrites are thus much less random than they would be if the directions were randomly drawn from a population described by equation (2). A fifth site (NZ31) for

which a confidence circle is not known appears to belong to this group. The simplest interpretation from the paleomagnetic point of view is that at least some of these petrologically distinct ignimbrites of this group were erupted within a time interval that probably was no more than a few centuries in length.

Finally the Marshall Ignimbrite (Site NZ12) has a different magnetic direction, designated "A" in Fig. 5. There can be no question that this direction is statistically different from the direction of ignimbrites in group "B" and that the Marshall Ignimbrite was formed at a different time.

Geologic information about the relative ages of these six ignimbrites from the Wanganui Series is summarised in Fig. 6. The stratigraphy shown in each vertical column is based on superposition or on the geologic field relationships of contiguous formations at the locality given at the top of the column. Data of this type provide firm information about the sequence of ignimbrite eruptions at each locality. Somewhat less secure is the correlation between different localities, because this is based in part on the assumption that petrologically similar ignimbrites are the same. Even less certain and more crucial to the problem of interpreting the magnetic results is the question of the amount of time represented by the unconformities. If, as seems likely, the time is more than a few thousand years, the magnetic directions of superimposed ignimbrites separated by an unconformity should in general be different.

A well-defined change in magnetic direction may be noted between the times of formation of the two oldest units sampled, the Marshall Ignimbrite (Site NZ12) and the Waiotapu Ignimbrite (Site NZ15). The two units are not directly superimposed, but the geologic information is at least consistent with an older age for the Marshall and the lapse of an interval of 10^3 years or more between the two. On the other hand, all of the younger ignimbrites of the Wanganui Series have nearly parallel directions. This was not expected in view of the geologic unconformity between the Waiotapu and Whakamaru Ignimbrites, that between the Whakamaru and Manunui Ignimbrites, and that between the Te Whaiti and Rangitaiki Ignimbrites. The discrepancy between the geologic and paleomagnetic data may be reconciled by one of the following alternative explanations. (1) The time intervals corresponding to the geologic unconformities are of the order of several centuries or less. (2) The time required for the formation of younger ignimbrites of the Wanganui Series was much longer than several centuries, possibly 10^5 years or more, yet the direction of the earth's field remained constant during all of this time. (3) The ignimbrites formed over a long time interval during which the field was changing, yet by coincidence the magnetic field returned to the same direction at the time of each eruption.

From what is known about rates of change in the direction of the earth's field at other localities, explanation (2) appears unlikely. Also unlikely, moreover, is explanation (3), for there is an extremely small statistical probability that the field would return to the same direction at each of four or five randomly spaced times, as previously discussed. From the paleomagnetic point of view, therefore, explanation (1) is preferred.

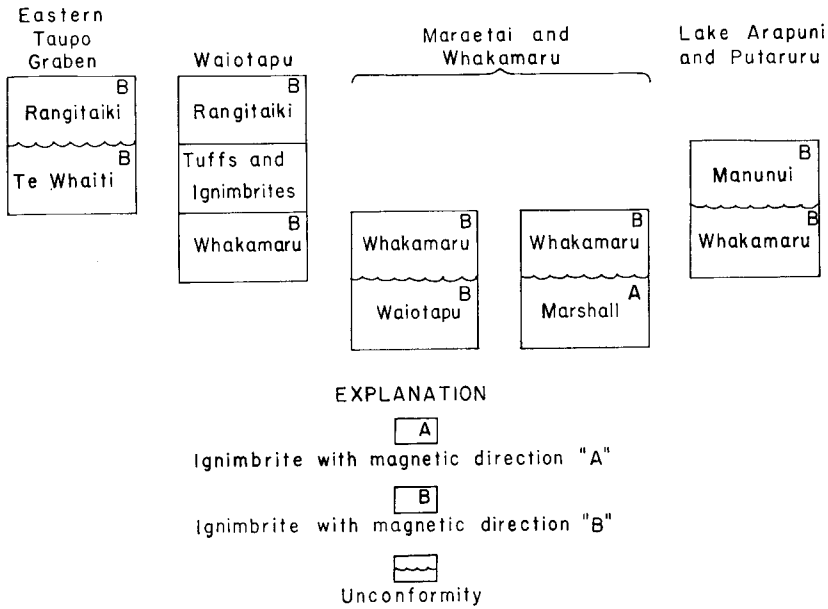


FIG. 6—Stratigraphic relationships of the six ignimbrites from the Wanganui Series which have magnetic directions "A" and "B" as given in Fig. 5. The locality where the stratigraphic relationship exists is indicated at the head of each column. The only ignimbrite with an "A" direction, the Marshall Ignimbrite, is probably older than the Waiotapu Ignimbrite. Based on geologic data from Grindley (1960), Healy *et al.* (1964), and in a personal communication from James Healy (1969).

Ignimbrites of the Wanganui Series younger than the Marshall Ignimbrite appear to have formed during one, or at most two, brief volcanic episodes when the earth's magnetic field in New Zealand was parallel to the mean direction of the "B" group.

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APPENDIX

EXPLANATION OF SITES

NOTE: Lower case letters such as om and mr refer to geologic map units in references (Grindley, 1960) and (Healy *et al.* 1964). Numbers such as N14/107367; P,44 are grid references based on the sheet districts of the 1:63,360 topographical map series (NZMS 1) and the national thousand-yard grid shown on that series. The edition and year of publication of the map sheet used are given after the grid reference, e.g., P,44 means provisional edition, 1944.

NZ1 Horeke Basalt (?) (N14/107367;P,44). Sampled along Taheke-Horeke Road 0·7 miles SE of McDonnel Road. Normally magnetized.

NZ2 Taheke Basalt (N14/163280;P,44). Cut on N side of main highway.

NZ3 Horeke Basalt (N15/337305;2,65). Sampled in quarry 2 miles S of Kaikohe on the E side of the airport road. Reversely magnetized.

NZ4 Horeke Basalt (N20/847016;3,64). Streambed at Whangarei Falls.

NZ5 Unmapped ignimbrite (N53/268927;3,65). Flow at least 15 m thick with massive columnar joints sampled in outcrop on W side of Waitekauri Road near railroad crossing 3·9 miles SW of Waihi on Paeroa–Waihi highway. Reversely magnetized.

NZ6 Omahia Andesite (om (Healy *et al.*, 1964)) (N53/414861;3,65). Quarry at junction of Waihi–Tauranga road with Athenree Road. Reversely magnetized.

NZ7 Minden Rhyolite (mr (Healey *et al.*, 1964)) (N58/526568;3,65). E side of the Minden Quarry. Reversely magnetized.

NZ8 Waitearíki Ignimbrite (N67/526448;2,65). Sampled in bed of river at McLarens Falls. Mapped as Mamaku Ignimbrite (Healy *et al.*, 1964) but J. Healy (pers. comm.) states that it is probably a much older unit, Waitearíki Ignimbrite. Reversely magnetized.

NZ9 Mamaku Ignimbrite (mk (Healy *et al.*, 1964)) (N67/647251;2,65). Quarry facing road between Ohaupara and Mangorewa Streams 300 m above their confluence.

NZ10 Ahuroa Ignimbrite member of Pakaumanu Group (paa (Healy *et al.*, 1964)) (NZ75/021119;2,67). Sampled in road cut 2.3 miles W of Pukeatua. Normally magnetized but rather unstable.

NZ11 Manunui Ignimbrite (mn (Healey *et al.*, 1964)) (NZ75/138130;2,67). Cut in road S of Arapuni Power Station.

NZ12 Marshall Ignimbrite (ms (Healy *et al.*, 1964)) (N75/323081;2,67). Small quarry on S side of road about 1 mile E of Lichfield.

NZ13 Haparangi Rhyolite (ha_1 (Healy *et al.*, 1964)) (N84/342694;1,64). Cliff outcrop along road on N side of Lake Whakamaru.

NZ14 Te Kopia Ignimbrite (te (Healy *et al.*, 1964)) (N85/733766;2,67). Sampled at base of precipitous rocky face forming a prominent fault scarp. Normal polarity.

NZ15 Waiotapu Ignimbrite (wp (Healy *et al.*, 1964)) (N85/799761;2,67). Cliff outcrop sampled near base. Trend of columnar joints suggests dip of 10° in azimuth 075°, but correction not made because of possibility that ignimbrite cooled against a sloping surface.

NZ16 Kaingaroa Ignimbrite (kg_2 (Healy *et al.*, 1964)) (N86/998720;2,68). Forestry Quarry on Webb Road. Samples collected near the middle of exposure which is 30 m thick.

NZ17 Tarawera Volcanic Complex (Cole, 1965) (N77/962928;2,65). Extreme western part of Ruawahia Dome is exposed on the N wall of the rift immediately E of the divide SW of the Ruawahia No. 2 Trig. Station.

NZ18 Matahina Ignimbrite (ma (Healy *et al.*, 1964)) (N86/222861;2,68). Shallow cut on Te Teko-Murupara Road.

NZ19 Tauhara Dacite (ta (Grindley, 1960)) (N94/638384;1,69). Mt Tauhara, quarry exposing dike leading to the W rim of a small cone on the E flank of the main volcano.

NZ20 Te Whaiti Ignimbrite (tw (Grindley, 1960)) (N104/934080;1,67). Cut in Napier-Taupo Road 3·0 miles W of Waipunga Stream.

NZ21 Rangitaiki Ignimbrite (ra (Grindley, 1960)) (N103/464144;1,64). Small cliff 100 m E of road adjacent to Hallets Bay, Lake Taupo.

NZ22 Pihanga Andesite (pi (Grindley, 1960)) (N102/260008;1,64). Cut on Ministry of Works road W of Papanetu Stream.

NZ23 North Crater Andesite (nc (Grindley, 1960)) (N112/140957;1,68). Large quarry NW of Otukou Village.

NZ24 Tongariro Andesite (to (Grindley, 1960)) (N112/038864;1,68). Exposures in cut on National Park - Taupo State Highway, 3 miles S of Taurewa.

NZ25 Ruapehu Andesite (ru (Grindley, 1960)) (N112/049718;1,68). Near base of series of thick flows which make up Te Herenga Ridge.

NZ26 Ruapehu Andesite (ru (Grindley, 1960)) (N112/030710;1,68). 1,200 yd S 45° W of Salt Hut.

NZ27 Ruapehu Andesite (ru (Grindley, 1960)) (N112/038717;1,68). Along bed of Tawhanui Stream, a tributary of Waikare Stream.

NZ28 Andesite flow of Mt Egmont (N119/688592;3,64). Streambed at Dawson Falls.

NZ29 Andesite flow of Mt Egmont (N119/667628;3,64). Second highest flow of series forming ridge below Warwick Castle.

NZ30 Andesite flow of Mt Egmont (N119/669619;3,64). East slope of Mt Egmont along trail S of ski area on N wall of Manganui Gorge.

NZ31 Whakamaru Ignimbrite. Normal magnetization reported (Hatherton, 1954) at three sites along the Waikato River: Waipapa, Maraetai, and the Whakamaru dam site. Not resampled for the present study.

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